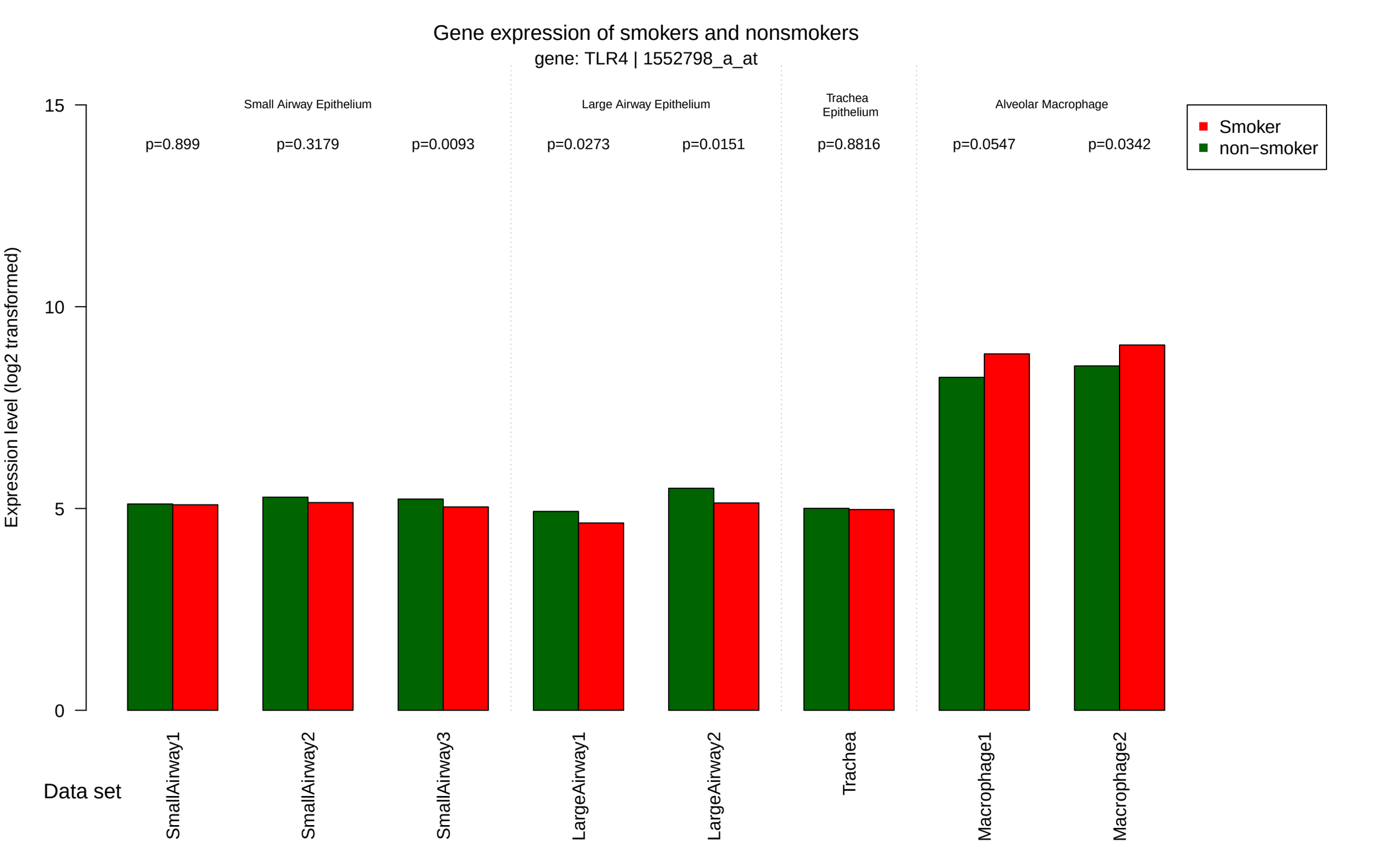
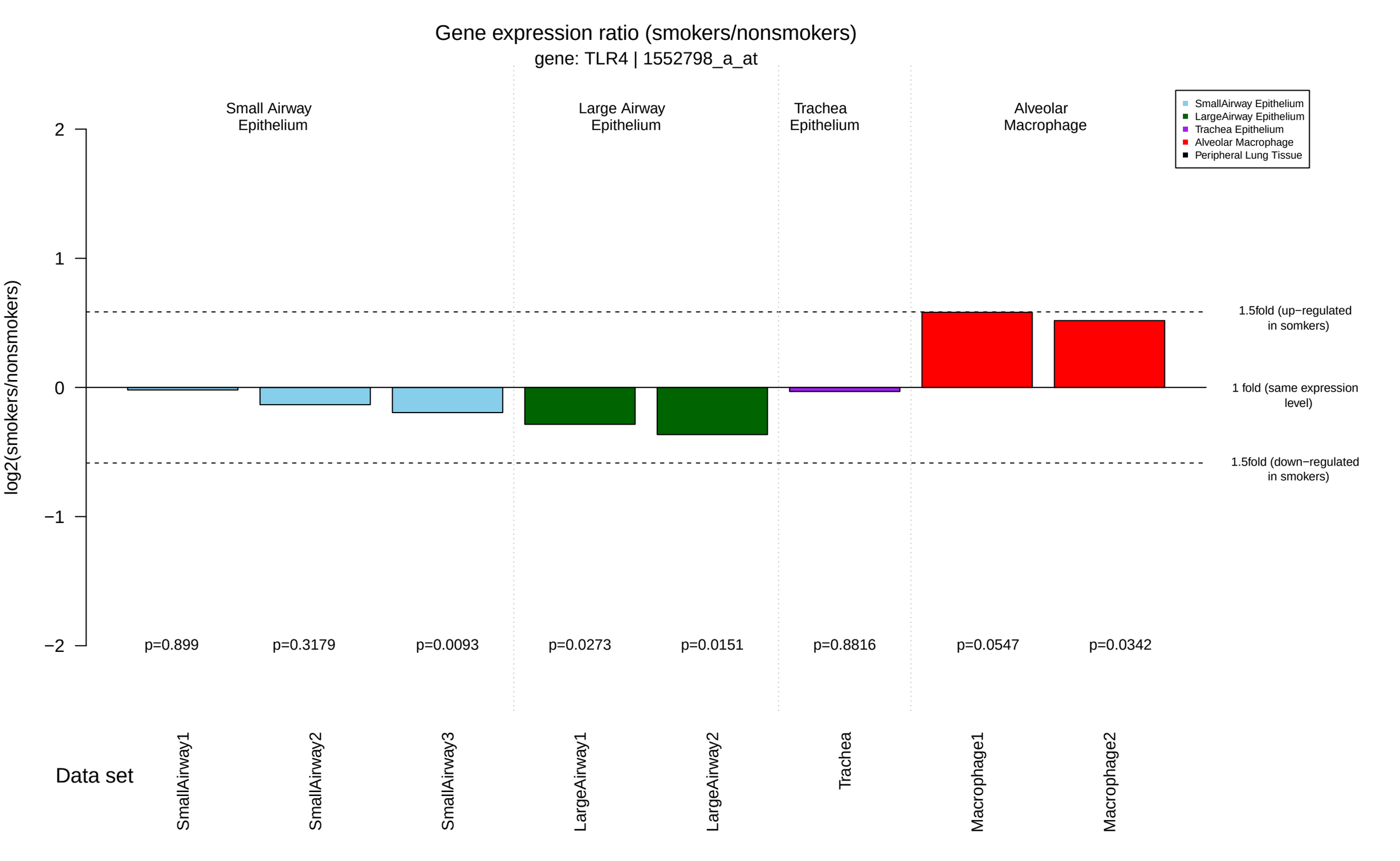
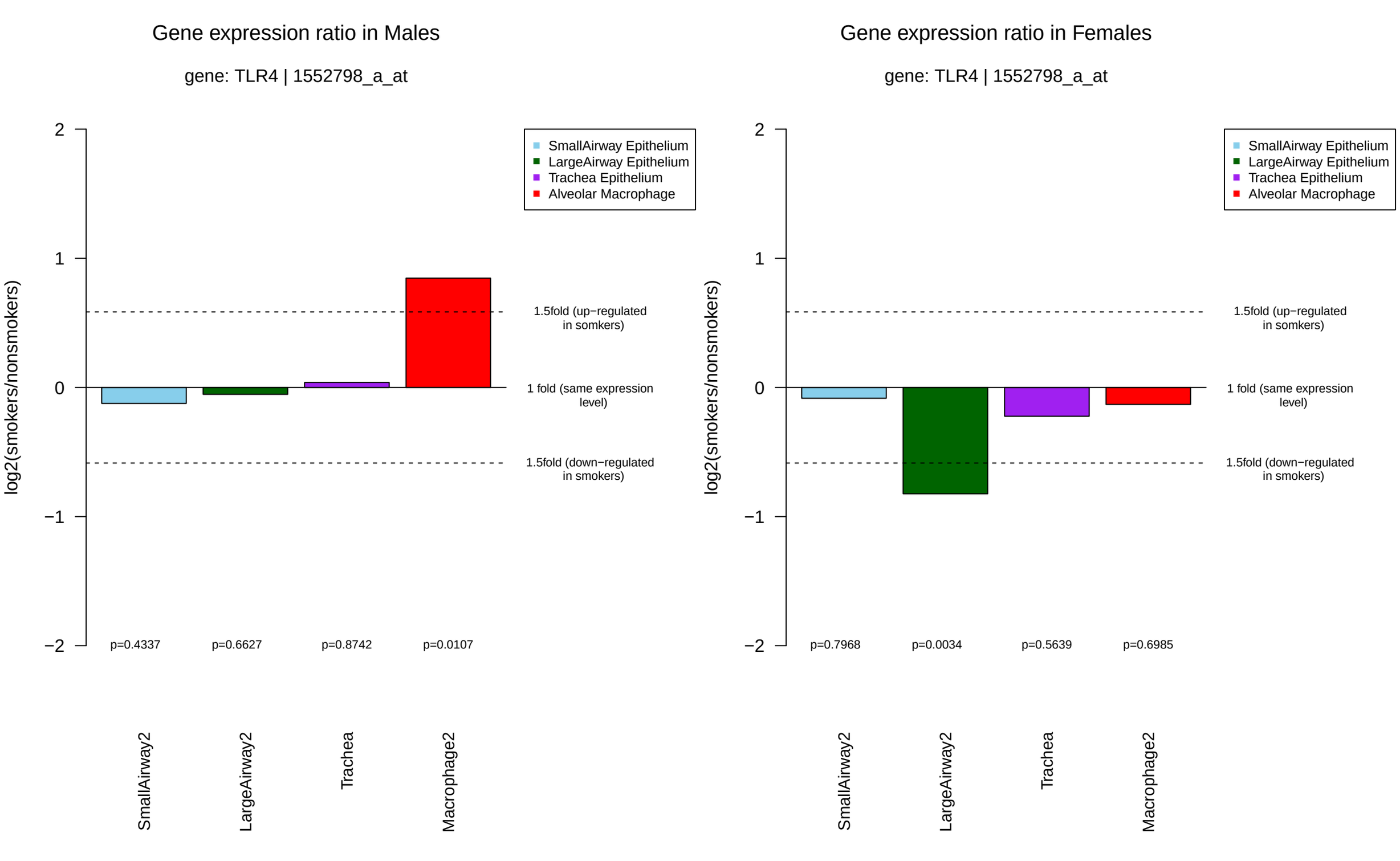
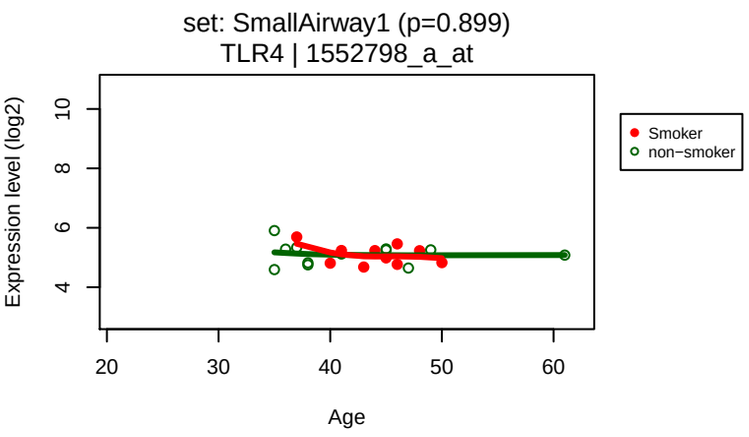
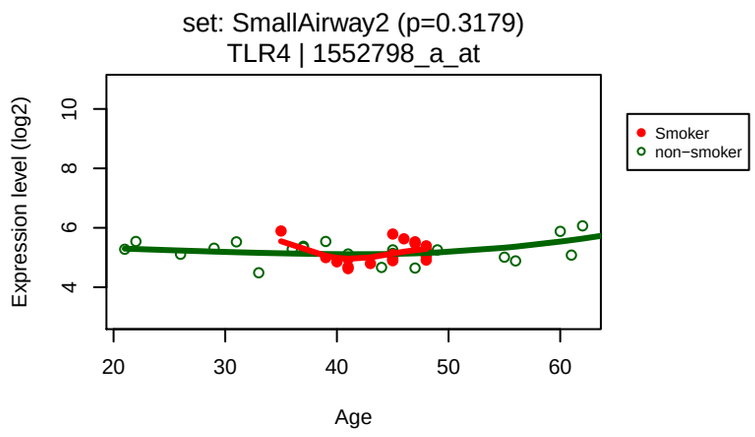
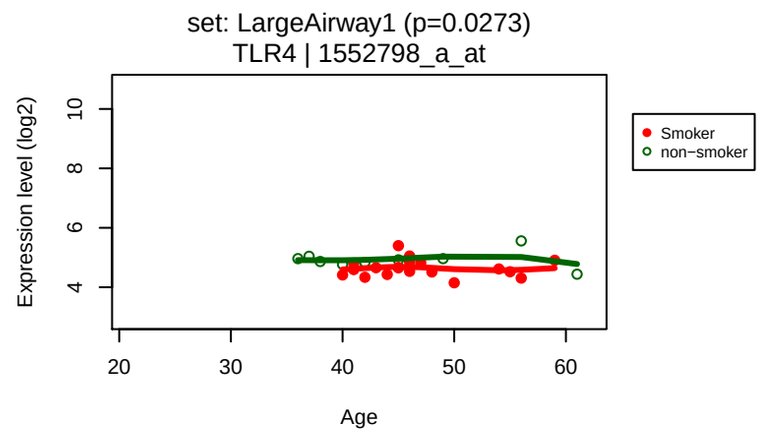

Gene expression of smokers and nonsmokers
gene: TLR4 | 1552798_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.899
p=0.3179
p=0.0093
p=0.0273
p=0.0151
p=0.8816
p=0.0547
p=0.0342
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

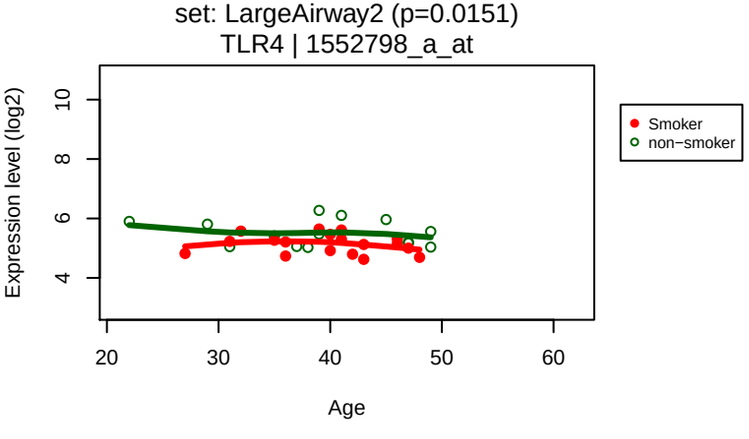

Gene expression ratio (smokers/nonsmokers)
gene: TLR4 | 1552798_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.899
p=0.3179
p=0.0093
p=0.0273
p=0.0151
p=0.8816
p=0.0547
p=0.0342
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

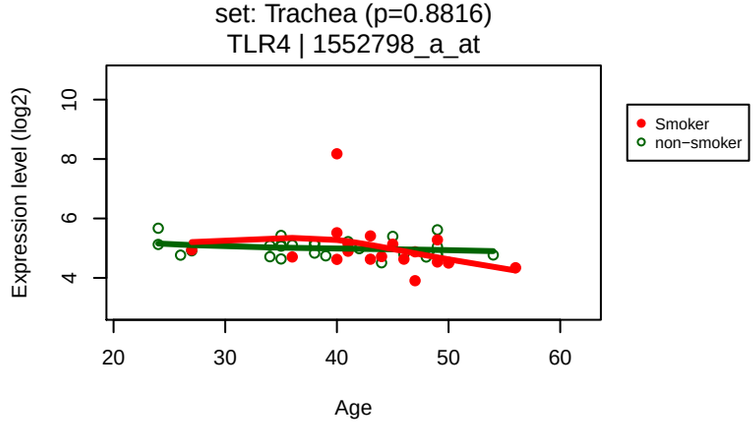

Gene expression ratio in Males
gene: TLR4 | 1552798_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4337
p=0.6627
p=0.8742
p=0.0107
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TLR4 | 1552798_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7968
p=0.0034
p=0.5639
p=0.6985
SmallAirway2
LargeAirway2
Trachea
Macrophage2

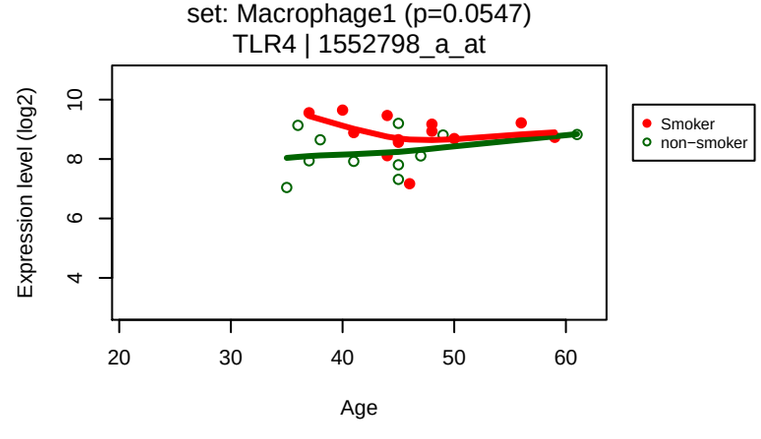

set: SmallAirway1 (p=0.899)
TLR4 | 1552798_a_at
Expression level (log2)
Age
Smoker
non−smoker

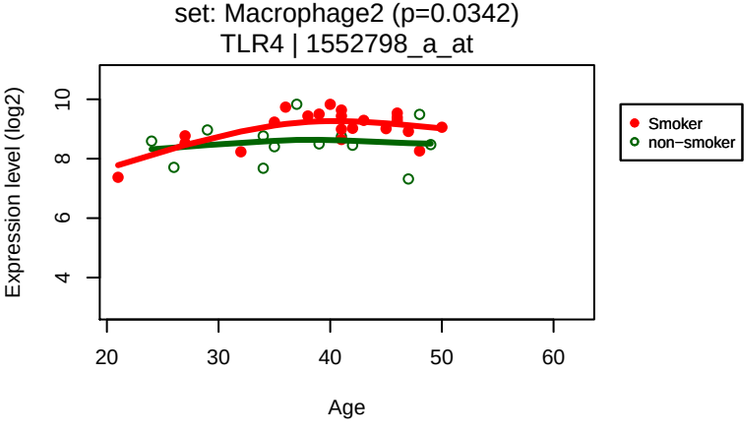

set: SmallAirway2 (p=0.3179)
TLR4 | 1552798_a_at
Expression level (log2)
Age
Smoker
non−smoker

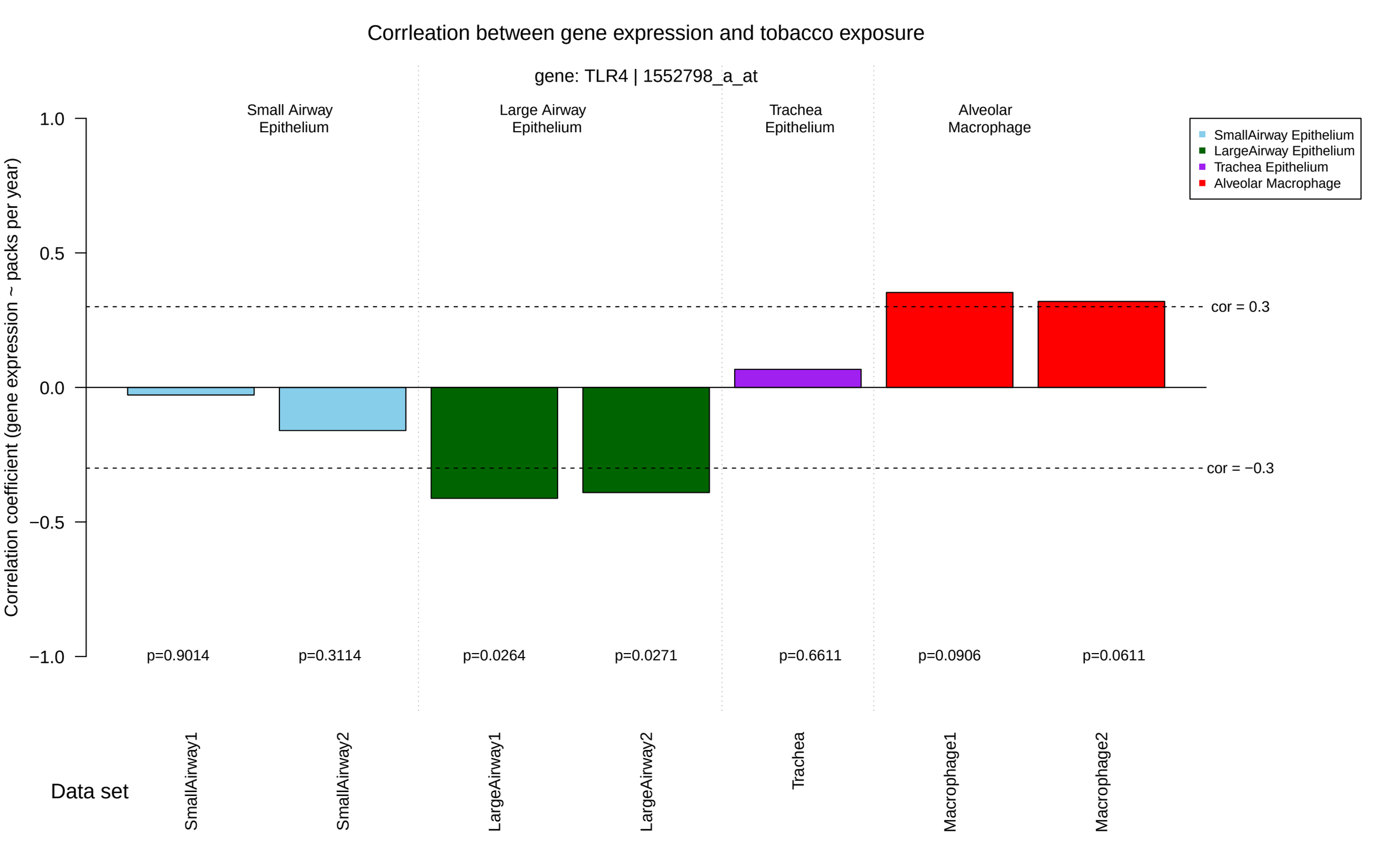

set: LargeAirway1 (p=0.0273)
TLR4 | 1552798_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0151)
TLR4 | 1552798_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8816)
TLR4 | 1552798_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0547)
TLR4 | 1552798_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0342)
TLR4 | 1552798_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TLR4 | 1552798_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9014
p=0.3114
p=0.0264
p=0.0271
p=0.6611
p=0.0906
p=0.0611
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2